Estimating probable maximum precipitation for Bangladesh

Sheikh Hefzul Bari

Department of Civil Engineering,
Leading University,
Sylhet-3112, Bangladesh
Email: shbari.bd@gmail.com

Abstract: In this paper, one-day probable maximum precipitation (PMP) is estimated for 29 meteorological stations of Bangladesh. A widely recognised statistical method, viz. Hershfield method, is used for PMP estimation. Afterward, the spatial prediction map of PMP is generated based on geostatistical analysis. It is found that Hershfield's frequency factor (K_m) for estimating PMP varies from 2.03 to 8.03 while most are around 3 to 4. Results indicate that higher values of PMP are found in the eastern part of the country whereas lower at the western part. However, maximum PMP (811 mm) is found in Bhola which is located in the coastal region. The second highest is also found in the coastal region. Results from the geostatistical analysis shows that empirical Bayesian kriging (EBK) performs slightly better in the spatial prediction map generation.

Keywords: probable maximum precipitation; PMP; Bangladesh; Hershfield method; empirical Bayesian Kriging; EBK; geostatistical analysis.

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Biographical notes: Sheikh Hefzul Bari is currently working as an Assistant Professor at Department of Civil Engineering at Leading University, Sylhet. His research interests are statistical hydrology, climate change analysis, GIS and remote sensing applications in environmental study, water resources, environmental science, etc.

1 Introduction

Probable maximum precipitation (PMP) is the maximum precipitation which is theoretically possible in a given time under modern meteorological conditions (WMO, 2009). However, a comparatively easy concept might be potential maximum precipitation (PMP) as such an amount is probable, but instead, is potentially possible (Kunkel et al., 2013). PMP is used for the estimation of the probable maximum flood (PMF) that is useful in the design of several hydraulic and flood-carrying structures. The concept of PMP is useful in the design of infrastructure for water retention and or routing, where failure would be catastrophic (Kunkel et al., 2013).

Bangladesh is a riverine country where water resources have a variety of status in various parts of the country. Consequently, hydraulics structures are of a variety of specifications depending on needs. For example, freshwater retaining structures are of main interest for the coastal and northwest regions. In contrast, flood control structures are very common in parts of the coastal region and northeastern low lying areas. The estimation of PMP plays an important role in either case. In addition, one-day PMP plays a vital role in stormwater drainage design in city areas. Therefore, the estimation of one-day PMP will provide a quick overview for designing any hydrological structures in Bangladesh. Still, despite being favourably important, the estimation of PMP for Bangladesh has stayed overlooked. Therefore, in this study, PMP has been estimated for Bangladesh and its spatial prediction map has been generated using the Kriging interpolation technique.

There are two major approaches available to estimate the value of PMP (Desa and Rakhecha, 2007), among which the statistical approach is quick as well as requires only precipitation data. This approach is preferred in those areas where meteorological parameters such as hourly precipitation, dew point temperature, and wind speed data are unavailable (Boota et al., 2015). There is no requirement of a meteorologist and this technique requires less time to apply (Boota et al., 2015; WMO, 2009). However, the statistical approach is used where adequate precipitation data (at least 30 to 40 years) are available (Chavan and Srinivas, 2015; WMO, 2009).

The most commonly used statistical method for estimating PMP is developed by Hershfield (1965). Although this is not the only statistical method, it has received the widest acceptance around the world and has become one of the standard methods suggested by the World Meteorological Organization (WMO) (Koutsoyiannis, 1999; WMO, 2009). It has the advantage of taking account of the actual historical data and being easy to use (Koutsoyiannis, 1999). The use of the Hershfield method has shown that the results obtained by this method are closely comparable to those obtained by the elaborate physical methods (Desa et al., 2001; Desa and Rakhecha, 2007).

2 Materials and methods

Daily rainfall data of 35 meteorological stations were collected from the Bangladesh Meteorological Department (BMD) for the time period 1948 to 2013. The collected data series were then checked for missing values. Following criterions were used for dealing with missing values:

a a year is considered missing if there are missing days between April to September

b if there are two or more consecutive missing years, they were discarded from the analysis

c daily maximum rainfall for a single year at any station was calculated from averaging previous and next year's daily maximum and compared with neighbouring stations for consistency.

After the treatment of missing values, PMP is calculated for all meteorological stations in Bangladesh having at least 30 years of rainfall data. There were 29 stations having rainfall data for more than 30 years and used in this study. Locations of the rainfall stations used in this study are given in Figure 1.

The Hershfield technique of estimating PMP is based on the general frequency equation (Chow, 1951). Hershfield (1961, 1965) proposed that PMP for a station can be estimated from the following equation:

$$X_{PMP} = \overline{X}_n + K_m \cdot \sigma_n$$

where X_{PMP} is the PMP for a given station for a specific duration and $\overline{X}_n$ and σ_n are the mean and standard deviation for a series of n annual daily maximum rainfall values respectively. K_m is the frequency factor.

K_m can be calculated using the following formula

$$K_m = \left(X_1 - \overline{X}_{n-1}\right) / \sigma_{n-1}$$

where

X_1 highest value in the series

$\overline{X}_{n-1}$ mean excluding highest value in the series

$\sigma_{(n-1)}$ standard deviation excluding highest value.

Values of K_m varies inversely with mean annual maximum rainfall at any station and the frequency factor (K_m) varies 3 to 14.5 (Desa et al., 2001; WMO, 2009). Hershfield adopted the highest value rounded to 15 for estimating PMP. Later, studies showed that the use of K_m = 15 for PMP calculation regardless of location can lead to excessive overestimation of PMP for a station with the high value of mean and annual maximum rainfall (Desa et al., 2001; Desa and Rakhecha, 2007). Therefore, in this study, PMP is calculated using station wise individual K_m values calculated using the above equation.

The spatial prediction map of PMP values is generated using the Kriging interpolation method. Suitable kriging method with best-fitted semivariogram was identified using geostatistical analysis (Hussain et al., 2016). In this study, ordinary kriging with three semivariogram models (spherical, exponential and Gaussian variogram) and empirical Bayesian kriging (EBK) were used for the geo-statistical analysis. Semivariogram fitting is automatic in EBK (Hussain et al., 2016). The best model was found by comparing with root mean square error (RMSE), root mean square standardised error (RMSSE) and average standard error (ASE). While comparing the efficiency of various kriging methods, RMSSE was chosen as the main cross-validation parameter. An RMSSE value closer to unity indicates a well-fitted semivariogram model (Hussain et al., 2016). Besides, RMSE and ASE should be equal or close enough for considering the model as satisfactory. In this study, the geostatistical analysis is performed using the ArcGIS software package.

3 Results and discussion

It is found that the frequency factor (K_m) varies from 2.03 to 8.03 (Figure 1) while most are around 3 to 4. Hershfield noted in his study that the value of K_m is 3.5 for 100 years of return period if the length of the record is 50 years (Hershfield, 1965). Therefore, K_m values obtained in this study can be said satisfactory. Individual K_m values for each station are then used to estimate PMP. The resulting PMP values are plotted in Figure 2.

The graph clearly highlights the variation of 1-day PMP values at different stations. The maximum PMP is found in Bhola district and the maximum value is 811 mm. The second highest PMP value is seen in Sandwip with a PMP value of 721 mm. Srimangal, the wettest place in the country has a one-day PMP of 663 mm. On the other hand, Barishal, Bogra, Jessore, Rajshahi, Rangpur, and Madaripur have the PMP values below 300 mm and the lowest value 262 mm is observed in the Rajshahi. It should be noted that Rajshahi is the driest region of Bangladesh. Therefore, a low value of mean annual maximum rainfall might be contributed towards a lower PMP. It is observed that the maximum PMP is found over the northeastern, parts of southern and the southeastern regions of Bangladesh. Though Sylhet is situated near Cherapunji of Shillong plateau which receives the highest rainfall in the world, it is interesting to note that the PMP obtained in this location is only 386 mm. However, this may indicate that the intensity of 1-day rainfall is low but rather uniform here.

Figure 1 Location of rain gauge stations and their respective frequency factors (K_m) (see online version for colours)

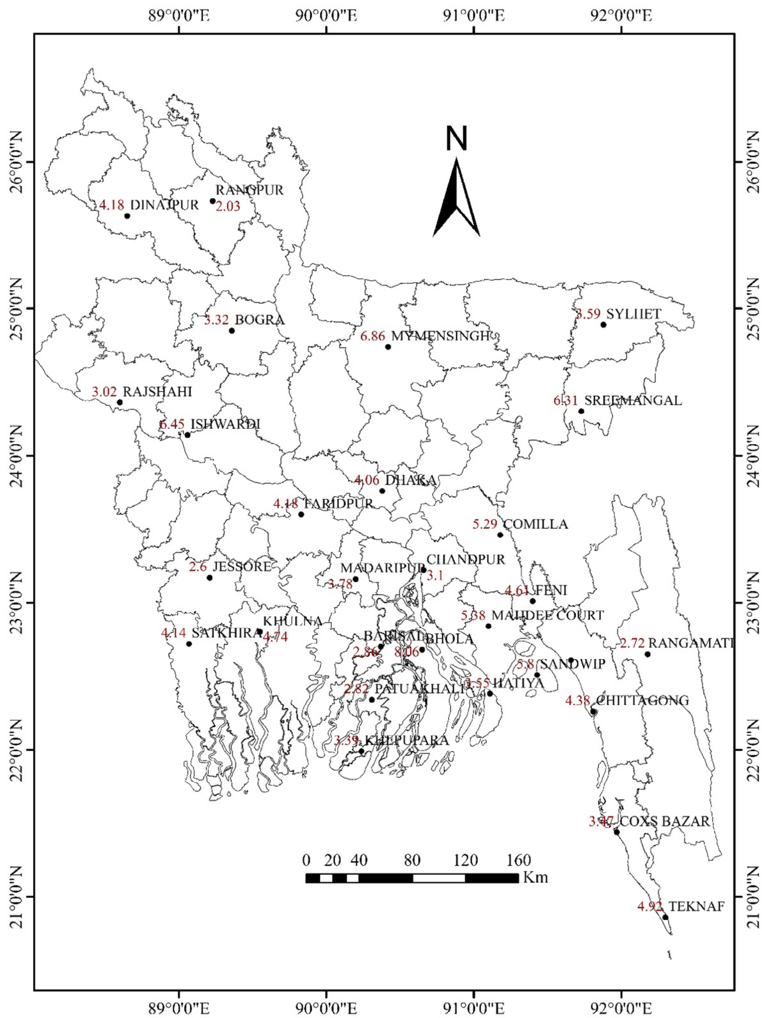
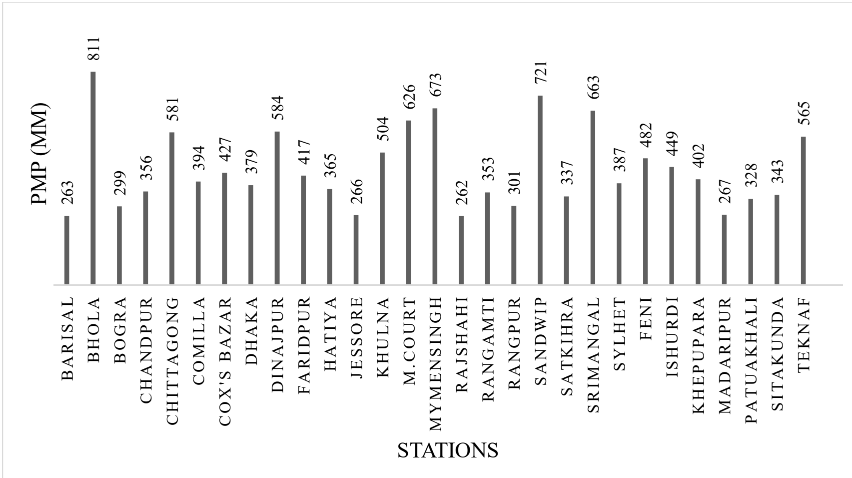


Figure 2 Estimated PMP values in rainfall stations



In order to determine the area PMP from point PMP values, geostatistical analysis with ordinary kriging using three semivariograms and EBK is applied. Performance of the ordinary kriging with three semivariograms and EBK is presented in Table 1. Results show that all ordinary kriging models gave the same results. However, overall results show that EBK would give the best performance. Therefore, the spatial prediction map is generated using EBK (Figure 3).

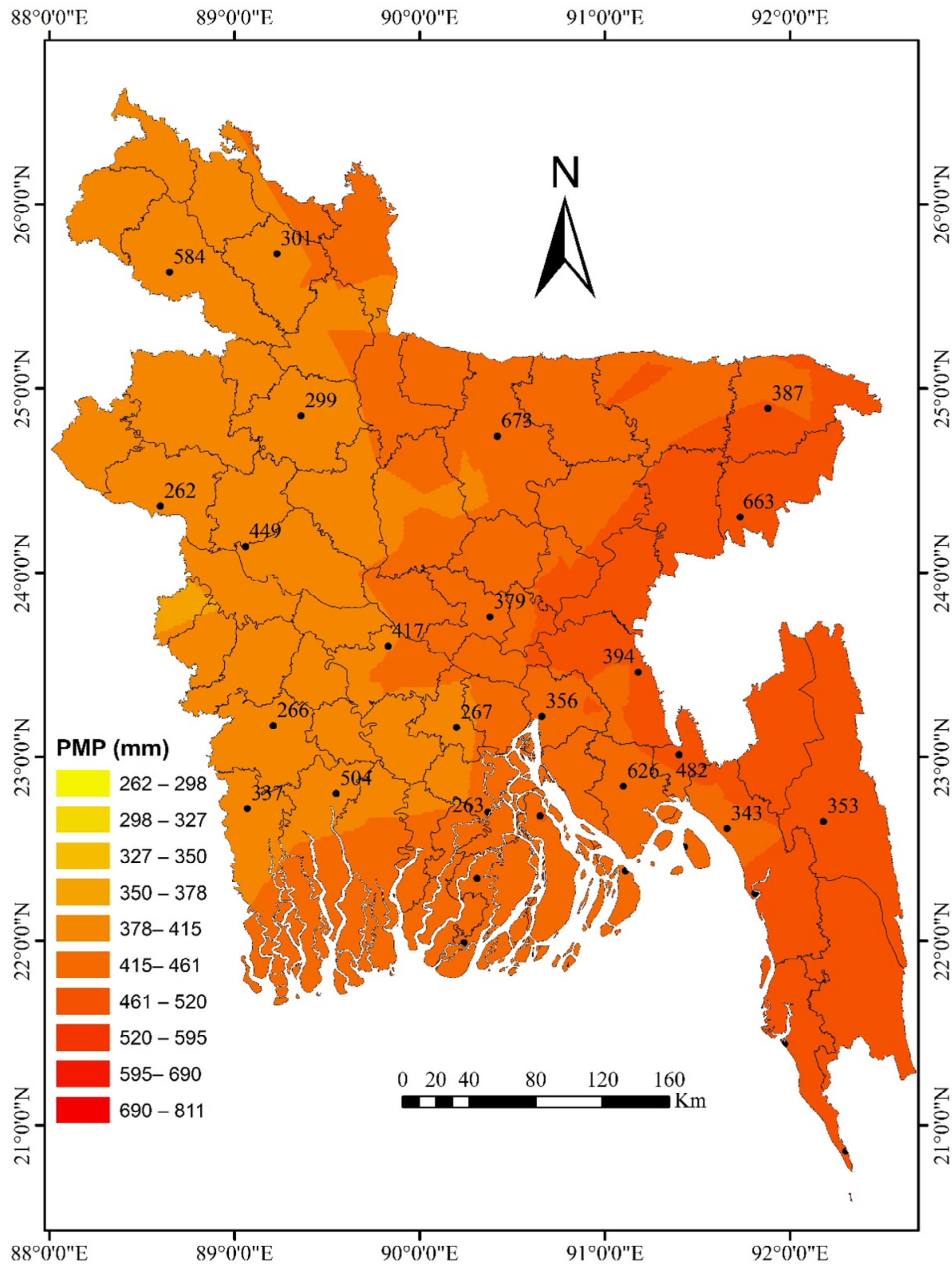
Table 1 Result of geostatistical analysis

		RMSE	*RMSSE*	*ASE*
Ordinary Kriging	Spherical	153.9131	0.9802415	160.7294
	Exponential	153.9131	0.9802415	160.7294
	Ordinary	153.9131	0.9802415	160.7294
Empirical Bayesian kriging (EBK)		*154.2551*	*0.9730147*	*158.8316*

4 Conclusions

An estimation of 1-day PMP using the Hershfield method for 29 meteorological stations in Bangladesh has been done in this study. The frequency factors are found varying between 2.03 to 8.03. It has been seen from the results that the northeastern and the southeastern hilly region have higher values of PMP. In contrast, the western region has lower PMP values. However, the top two PMP values are found in the coastal region. The highest PMP is found at Bhola (821 mm) and lowest at Barisal (263 mm). Average 1-day PMP over the country is found more than 400 mm. The spatial prediction map from point PMP values has been generated based on geostatistical analysis. Results of the geostatistical analysis show that all of the used technique performs almost similarly but EBK is found slightly better performing.

Figure 3 Spatial distribution of PMP over Bangladesh (see online version for colours)



It is anticipated that the estimated 1-day PMP will provide crude information to the respective authority regarding the planning and design of various hydraulic constructions. These findings can also be successfully used to investigate the adequacy of the hydraulic structures under extreme flood conditions and to estimate PMF.

References

Boota, M.W., Nabi, G., Ahsan, H. and Abbas, T. (2015) 'Estimation of probable maximum precipitation for one day duration: a case study of Gujjar Khan in Potwar region, Pakistan', *J. Himal. Earth Sci*., Vol. 48, No. 2, pp.148–154.

Chavan, S.R. and Srinivas, V.V. (2015) 'Probable maximum precipitation estimation for catchments in Mahanadi River Basin', *Aquat. Procedia*, Vol. 4, pp.892–899, https://doi.org/10.1016/j.aqpro.2015.02.112.

Chow, V.T. (1951) 'A general formula for hydrologic frequency analysis', *Eos Trans. Am. Geophys. Union*, Vol. 32, No. 2, pp.231–237, https://doi.org/10.1029/TR032i002p00231.

Desa, M.M.N. and Rakhecha, P.R. (2007) 'Probable maximum precipitation for 24-h duration over an equatorial region: part 2-Johor, Malaysia', *Atmospheric Res*., Vol. 84, pp.84–90, https://doi.org/10.1016/j.atmosres.2006.06.005.

Desa, M.M.N., Noriah, A.B. and Rakhecha, P.R. (2001) 'Probable maximum precipitation for 24 h duration over southeast Asian monsoon region – Selangor, Malaysia', *Atmospheric Res*., Vol. 58, pp.41–54, https://doi.org/10.1016/S0169-8095(01)00070-9.

Hershfield, D.M. (1961) 'Estimating the probable maximum precipitation', *J. Hydraulics Div. (Journal of the Hydraulics Division) Am. Soc. Civil Engrs*., Vol. 87, No. 5, pp.99–116, American Society of Civil Engineers.

Hershfield, D.M. (1965) 'Method for estimating probable maximum rainfall', *J. Am. Water Works Assoc*., Vol. 57, No. 8, pp.965–972.

Hussain, Md.M., Bari, S.H., Tarif, M.E., Rahman, Md.T.U. and Hoque, M.A. (2016) 'Temporal and spatial variation of groundwater level in Mymensingh district, Bangladesh', *Int. J. Hydrol. Sci. Technol*., Vol. 6, p.188, https://doi.org/10.1504/IJHST.2016.075587.

Koutsoyiannis, D. (1999) 'A probabilistic view of Hershfield's method for estimating probable maximum precipitation', *Water Resour. Res*., Vol. 35, pp.1313–1322, https://doi.org/10.1029/1999WR900002.

Kunkel, K.E., Karl, T.R., Easterling, D.R., Redmond, K., Young, J., Yin, X. and Hennon, P. (2013) 'Probable maximum precipitation and climate change', *Geophys. Res. Lett*., Vol. 40, pp.1402–1408, https://doi.org/10.1002/grl.50334.

World Meteorological Organization (WMO) (2009) *Manual on Estimation of Probable Maximum Precipitation (PMP)*, WMO-No.1045, Geneva, Switzerland.